DD2434/FDD3434 Machine Learning, Advanced Course
Assignment 1E, 2026

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Deadline, see Canvas

Read this before starting

You will present the assignment by a written report in PDF format, submitted before the deadline using Canvas. The assignment should be done in groups of two, and it will automatically be checked for similarities to other students' solutions as well as documents on the web in general. Although you are allowed to discuss the problem formulations with other groups, you are not allowed to discuss solutions, and any discussions concerning the problem formulations must be described in the solutions you hand in (including which group you discussed with).

From the report it should be clear what you have done and you need to support your claims with results. You are supposed to write down the answers to the specific questions detailed for each task. This report should clearly show how you have drawn your conclusions and explain your derivations. Your assumptions, if any, should be stated clearly. Show the results of your experiments using images and graphs together with your analysis and add the commented version of your code as an appendix.

Code submission. In addition to the report, submit your completed notebooks and any supporting source files for *both* parts of Assignment 1E (Bayesian linear regression and CAVI) in two clearly labelled versions: **one with comments and one without comments**. Both versions must contain the same executable code and produce the same results; remove only comments from the second version. We will use the **version without comments during the oral exam**, where you will be asked to explain and discuss your implementation.

Being able to communicate results and conclusions is a key aspect of scientific as well as corporate activities. It is up to you as an author to make sure that the report clearly shows what you have done. Based on this, and only this, we will decide if you pass the task. No detective work should be required on our side. In particular, neat and tidy reports please!

The grading of the assignment 1E, 2E and 3E (20 points each) will be as follows,

E 45 points, with at least 10 points from each Assignment.

Good Luck!

1.1 Bayesian Linear Regression

For inputs $x_i \in \mathbb{R}$, let $\tilde{\boldsymbol{x}}_i = (1, x_i)^\top$ be the design vector (bias and slope) and $\boldsymbol{w} = (w_0, w_1)^\top$ the weights. The model is

$$y_i = \tilde{\boldsymbol{x}}_i^\top \boldsymbol{w} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \qquad \boldsymbol{w} \sim \mathcal{N}(\boldsymbol{m}_0, \boldsymbol{S}_0),$$

where the noise variance σ^2 is *known*. Stack the design vectors in $X \in \mathbb{R}^{N \times 2}$. As discussed in the Module 1 exercise session, Exercise 1.5, the posterior and the posterior predictive distribution at a new input $x_\star$ are given by the following expressions:

$$p(\boldsymbol{w} \mid X, \boldsymbol{y}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{m}_N, \boldsymbol{S}_N), \quad \boldsymbol{S}_N = \big(\boldsymbol{S}_0^{-1} + \sigma^{-2} X^\top X\big)^{-1}, \quad \boldsymbol{m}_N = \boldsymbol{S}_N\big(\boldsymbol{S}_0^{-1}\boldsymbol{m}_0 + \sigma^{-2} X^\top \boldsymbol{y}\big),$$
$$p(y_\star \mid x_\star, X, \boldsymbol{y}) = \mathcal{N}\big(y_\star \mid \tilde{\boldsymbol{x}}_\star^\top \boldsymbol{m}_N,\ \sigma^2 + \tilde{\boldsymbol{x}}_\star^\top \boldsymbol{S}_N \tilde{\boldsymbol{x}}_\star\big),$$

where $\tilde{\boldsymbol{x}}_\star = (1, x_\star)^\top$. Using the maximum-likelihood weights instead gives the ML predictive distribution $\mathcal{N}\big(\tilde{\boldsymbol{x}}_\star^\top \hat{\boldsymbol{w}}^{\text{ML}}, \sigma^2\big)$.

Use the template notebook `1E-BLR-HT26.ipynb` provided on Canvas. The notebook contains an implementation of the data-simulation procedure, together with functions for loading the penguin data and computing the closed-form posterior, the ML estimate, and the posterior predictive distribution. **Solutions that do not use the template will not receive any points.** Use numpy and scipy for the implementation (do not use torch), and do not change the seeds in the notebook. For the simulated data (Questions 1.1.1 and 1.1.4), $\sigma = 0.5$, $\boldsymbol{m}_0 = \boldsymbol{0}$, $\boldsymbol{S}_0 = I$, and the training inputs are drawn as $x_i \sim U(-1, 1)$. In the report, include the code of every function you implement.

Question 1.1.1: ***Checking the posterior.***

(a) *Implement a function that evaluates the unnormalised log posterior* $\log p(\boldsymbol{y} \mid X, \boldsymbol{w}) + \log p(\boldsymbol{w})$ *on a grid over* (w_0, w_1)*, using only log-densities of the likelihood and the prior (not the closed-form posterior), and normalises it numerically on the grid. For* $N = 2, 10, 100$*, overlay contour plots of the grid posterior and the closed-form posterior, and report the maximum absolute difference between the two densities on the grid.*

(b) *Implement sequential Bayesian updating, where the posterior after* n *data points is used as the prior for data point* $n + 1$*. Show numerically that, after processing all* $N = 100$ *observations, the sequential posterior agrees with the batch posterior up to numerical precision. Compare their posterior mean vectors and covariance matrices. Plot the posterior after* $n = 0, 1, 2, 5, 20, 100$ *points with the true weights marked, and briefly comment on how the posterior changes as* n *grows.*

(2 points)

Real data. Questions 1.1.2 and 1.1.3 use the Palmer penguins dataset (Horst, Hill & Gorman, 2020), provided as `data/penguins.csv`: flipper length (input x) and body mass (output y) of 342 penguins of three species, both standardised. Each observation is a pair of measurements from one penguin. In the notebook, you can find code that estimates the noise standard deviation σ from the residuals of a least-squares fit to the full penguin dataset. Use this estimate as a fixed, known value in the subsequent real-data experiments. The provided code also specifies the prior $\mathcal{N}(\mathbf{0}, I)$ and defines a *training pool* (the 70% of penguins with the shortest flippers) and an *extrapolation region* (the remaining 30%).

Question 1.1.2: ***Uncertainty and extrapolation on real data.*** *For each supplied training set of size $N = 2, 10, 100$, drawn from the training pool, implement* `predictive_decomposition` *to compute the aleatoric contribution σ^2 and the epistemic contribution $\tilde{\boldsymbol{x}}_\star^\top \boldsymbol{S}_N \tilde{\boldsymbol{x}}_\star$ to the posterior predictive variance at inputs on a grid spanning the full observed range of flipper lengths. Use the posterior covariance $\boldsymbol{S}_N$ obtained from that training set.*

Produce one panel for each N, showing:

- *A scatter plot of all 342 observations, with standardised flipper length on the horizontal axis and standardised body mass on the vertical axis. Each point represents one penguin. Highlight the N observations used for training using a distinct colour or marker.*
- *The posterior predictive mean with its 95% predictive interval, based on the total variance $\sigma^2 + \tilde{\boldsymbol{x}}_\star^\top \boldsymbol{S}_N \tilde{\boldsymbol{x}}_\star$.*
- *The pointwise 95% posterior credible interval for the noise-free regression mean:*

$$\tilde{\boldsymbol{x}}_\star^\top \boldsymbol{m}_N \;\pm\; 1.96\sqrt{\tilde{\boldsymbol{x}}_\star^\top \boldsymbol{S}_N \tilde{\boldsymbol{x}}_\star}.$$

  This interval describes uncertainty about the regression mean and does not include observation noise.

- *The ML prediction with its 95% interval $\pm 1.96\,\sigma$.*

Compare the widths of the three plotted intervals within the training-pool input range and in the extrapolation region. Explain how these widths change as N increases. For the Bayesian and ML predictive intervals, discuss how well they include the observed body masses in the extrapolation region. (2 points)

Question 1.1.3: ***Calibration of predictive intervals on real data.*** *A 95% predictive interval is calibrated if it contains a new observation 95% of the time. Implement a resampling experiment that, in each of $M = 3000$ replicates, draws N training observations, draws one test observation not in the training set, and records whether its body mass lies inside (i) the Bayesian 95% predictive interval and (ii) the ML 95% interval. Run it for $N \in \{2, 10, 100\}$ and two designs:* random*, where training and test observations are drawn from the full penguin dataset, and* extrapolation*, where training observations come from the training pool and test observations from the extrapolation region.*

For each training-set size and sampling design, report the percentage of the $M = 3000$ repetitions in which the held-out test observation falls within the Bayesian predictive interval, and the corresponding percentage for the ML predictive interval. These percentages are the estimated empirical coverages*. Present the results in a table with columns for sampling design, N, Bayesian coverage (%), and ML coverage (%). Explain when the Bayesian interval is approximately calibrated, when and why the ML interval fails, and why the Bayesian coverage in the extrapolation design gets worse as N grows. (3 points)*

Question 1.1.4: ***When the prior is wrong.*** *We return to simulated data ($\sigma = 0.5$, $x_i \sim U(-1, 1)$), where the true weights are known. Implement a Monte Carlo experiment that, in each of $M = 5000$ replicates, draws true weights $\boldsymbol{w} \sim \mathcal{N}\big((2, 2)^\top, I\big)$, simulates a training set of size N, draws one new observation $y_\star$ at $x_\star = 3$, and records whether $y_\star$ lies inside the Bayesian 95% predictive interval computed with the prior $\mathcal{N}(\mathbf{0}, s_0^2 I)$, and inside the ML 95% interval. Report the estimated empirical coverages for the Bayesian intervals with $s_0^2 \in \{0.1, 1, 10\}$ and $N \in \{2, 10, 100\}$, together with those for the ML intervals, in a table. As in Question 1.1.3, empirical coverage is the percentage of repetitions in which the interval contains the new observation; use $M = 5000$ repetitions here. Discuss how the confidence of the prior (s_0^2) and the mismatch between the prior mean and the true weights interact with N. In which settings is the Bayesian predictive worse calibrated than the ML predictive, and why? Why does this problem disappear as N grows, unlike the model misspecification in Question 1.1.3? The notebook also contains an optional, ungraded extension on choosing s_0^2 with the marginal likelihood. (3 points)*

1.2 CAVI

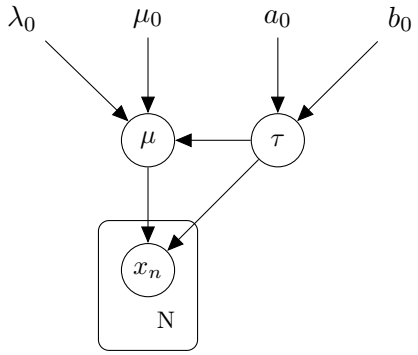


Figure 1: Bayesian network of the Normal-NormalGamma model.

Consider the model defined by Equations (10.21)–(10.23) in Bishop, whose DGM is shown in Figure 1. This question concerns the VI algorithm for this model that was covered in the lectures and in the book. Unlike in Section 1.1, where the noise variance was known, the precision τ of the Gaussian is now unknown as well and is inferred together with the mean μ.

Use the template notebook `1E-CAVI-HT26.ipynb` provided on Canvas. **Solutions that do not use the template will not receive any points.** Use numpy and scipy for the implementation (do not use torch), and do not change the seeds in the notebook.

Question 1.2.5: *Implement a function that generates data points for the given model. Set $\mu = 1$, $\tau = 0.5$ and generate datasets of size $N = 10, 100, 1000$. Plot a histogram of each of the 3 datasets. (1 points)*

Question 1.2.6: *Write out the expressions for the ML estimates of μ and τ (without derivations). Implement a function that computes and returns the ML estimates for given data. (1 points)*

Question 1.2.7: *What is the exact posterior? (Show your derivations.) (2 points)*

Question 1.2.8: *Implement the VI algorithm for the variational distribution in Equation (10.24) in Bishop. Run the VI algorithm on the datasets. Plot the ELBO over iterations. Compare the inferred variational distribution with the exact posterior and the ML estimate. Visualize the results and discuss your findings. (6 points)*